autotests in the acceptance stage of the deployment pipeline (as explained in chapter 3), to test applications deployed in a production-like environment.

Polar Labs

Until now, the only application supposed to be accessed by users directly was Edge Service. All the other Spring Boot applications interact with each other from within the environment where they are deployed.

Service-to-service interactions within the same Docker network or Kubernetes cluster can be configured using the container name or the Service name respectively. For example, Edge Service forwards requests to Polar UI via the `http://polar-ui:9004` URL on Docker (`<container-name>:<container-port>`) and via the `http://polar-ui` URL on Kubernetes (Service name).

Keycloak is different because it's involved in service-to-service interactions (for now, those are just interactions with Edge Service) and also interactions with end users via the web browser. In production, Keycloak will be accessible via a public URL that both applications and users will use, so there will be no problem. How about in local environments?

Since we don't deal with public URLs when working locally, we need to configure things differently. On Docker, we can solve the problem by using the `http://host.docker.internal` special URL configured automatically when installing the software. It resolves to your localhost IP address and can be used both within a Docker network and outside.

On Kubernetes, we don't have a generic URL to let Pods within a cluster access your local host. That means Edge Service will interact with Keycloak via its Service name (`http://polar-keycloak`). When Spring Security redirects a user to Keycloak to log in, the browser will return an error because the `http://polar-keycloak` URL cannot be resolved outside the cluster. To make that possible, we can update the local DNS configuration to resolve the `polar-keycloak` hostname to the cluster IP address. Then a dedicated Ingress will make it possible to access Keycloak when requests are directed to the `polar-keycloak` hostname.

If you're on Linux or macOS, you can map the `polar-keycloak` hostname to the minikube local IP address in the /etc/hosts file. On Linux, the IP address is the one returned by